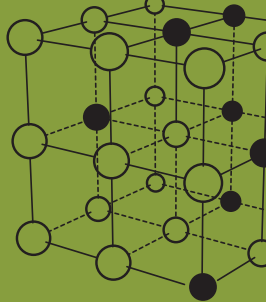




MELBA ROY





Melba Roy Mouton (b. 1929)
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Mouton worked at NASA's
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illustration: Kristin DiVona



Section 1.3-Quadratic Equations

DEFINITION Quadratic Equation

A **quadratic equation** is an equation equivalent to one of the form

$$ax^2 + bx + c = 0 \quad a \neq 0 \quad (1)$$

where a , b , and c are real numbers.

A quadratic equation of the form $ax^2 + bx + c = 0$ is in standard form and can be solved for the unknown variable by using one of the 5 methods below:

5 Methods for Solving a Quadratic Equation	
1.	Solve by Factoring
2.	Solve by Using the Square Root Method
3.	Solve by Completing the Square
4.	Solve by using the Quadratic Formula
5.	Solve Graphically by using Desmos or other Graphing Technology

Ex 1. Solve the following quadratic equations by factoring.

a) $x^2 - 9x = 0$

Need to Review?

Factoring is discussed in
Section R.5, pp. 51-57.

b) $z^2 + z - 6 = 0$

Ex 2. Use the **square root method** to solve the following quadratic equations:

a) $(x-1)^2 = 4$

b) $x^2 - 16 = 0$

Need to Review?

- Completing the square is discussed in Section R.5, pp. 57–58.

Completing the Square Method

Process for Completing the Square Given a Quadratic Equation in Standard Form $ax^2 + bx + c = 0$

Step 1: First move any constant term to the other side of the equal sign.

Step 2: Divide both sides of the equation by the variable a in front of the squared term.

Step 3: Determine which 'magic' number to add to both sides of the equation by finding $(\frac{b}{2})^2$ and adding it to both sides of the equation.

Step 4: Factor the left hand side into a binomial squared and simplify the right hand side. The left hand side should factor into $(x \pm \frac{b}{2})^2$

Ex 3. Use the **completing the square method** to solve the following quadratic equations:

a) $x^2 + 6x + 1 = 0$

NOTE If the coefficient of the square term is not 1, divide both sides by the coefficient of the square term before attempting to complete the square. ■

b) $2x^2 - 3x - 1 = 0$

THEOREM Quadratic Formula

Consider the quadratic equation

$$ax^2 + bx + c = 0 \quad a \neq 0$$

- If $b^2 - 4ac < 0$, this equation has no real solution.
- If $b^2 - 4ac \geq 0$, the real solution(s) of this equation is (are) given by the **quadratic formula**:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (5)$$

The **Discriminant** of a Quadratic Equation is $b^2 - 4ac$

Discriminant of a Quadratic Equation

For a quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$:

- If $b^2 - 4ac > 0$, there are two unequal real solutions.
- If $b^2 - 4ac = 0$, there is a repeated real solution, a double root.
- If $b^2 - 4ac < 0$, there is no real solution.

Ex 4. First use the Discriminant to determine how many real solutions the following quadratic equations will have. Then find any real solutions that exist using the Quadratic Formula.

a) $4t^2 + t + 1 = 0$

b) $2x^2 + 5x + 3 = 0$

Ex 5. Use the Graphical Method to find all solutions to the following quadratic equations:

a) $x^2 - 4.1x + 2.2 = 0$

b) $2x^2 = -x - 7$